Comparative Evaluation of Space Mean Speed Estimation Methods Using Trajectory Data and Edie's Definition

Prepared by: Saurav Barua

Work Plan

Location: Henri-Dunant Intersection

Methods:

Method 1: Entry-Exit travel time-based Space Mean Speed

Method 2: Instantaneous speed-based Space Mean Speed

Method 3: Trajectory based aggregation of Edie's Space Mean Speed

Method 4: Entry-Based aggregated Count Method for Space Mean Speed Estimation

Screen shot from video showing entry and exit gate

Method 1: Entry-Exit travel time-based Space Mean Speed

✅ Method 1: Entry–Exit Travel Time Based Space Mean Speed

🔹 Concept

Space Mean Speed (SMS) is the **harmonic mean of vehicle speeds**, computed using **trip-level data (distance & travel time per vehicle)**.

📄 Formula

$$v_i = \frac{d_i}{t_i}$$

$$SMS = \frac{N}{\sum_{i=1}^{N} \frac{1}{v_i}}$$

Where:

- d_i = traveled distance of vehicle i
- t_i = travel time of vehicle i
- v_i = average speed of vehicle i
- N = number of vehicles

Pseudo Code:

INPUT: vehicle-level dataset (Entry Time, Exit Time, Distance)

STEP 1: Remove duplicate rows → keep one row per vehicle

STEP 2: For each vehicle i:
travel_time[i] = Exit_Time[i] - Entry_Time[i]

STEP 3: Remove vehicles where:
travel_time[i] <= 0

STEP 4: Compute speed per vehicle:
speed[i] = distance[i] / travel_time[i]

STEP 5: Convert speed to km/h if needed:
speed[i] = speed[i] * 3.6

STEP 6: Compute Space Mean Speed (harmonic mean):
SMS = N / SUM(1 / speed[i])

OUTPUT: SMS

Method 2: Instantaneous speed-based Space Mean Speed

✅ Method 2: Distance-Weighted Time-Series Based Space Mean Speed (Correct)

🔹 Concept

This is the **true continuous formulation** of space mean speed using trajectory/time-series data.

It is equivalent to:

$$SMS = \frac{\text{Total Distance}}{\text{Total Travel Time}}$$

📄 Formula

For each time step:

$$\Delta d = v \cdot \Delta t$$

For each vehicle:

$$d_i = \sum \Delta d \quad , \quad t_i = \sum \Delta t$$

Overall:

$$SMS = \frac{\sum_i d_i}{\sum_i t_i}$$

Pseudo Code

INPUT: trajectory dataset (Track ID, Time, Speed)

STEP 1: Sort data by:
Track ID, Time

STEP 2: For each vehicle:
Compute time difference:
dt = current_time - previous_time

STEP 3: Remove rows where:
dt is NULL or dt <= 0

STEP 4: Convert speed:
speed_mps = speed_kmh / 3.6

STEP 5: Compute step distance:
step_distance = speed_mps * dt

STEP 6: Aggregate per vehicle:
total_distance[i] = SUM(step_distance)
total_time[i] = SUM(dt)

STEP 7: Compute per vehicle speed:
speed[i] = total_distance[i] / total_time[i]

STEP 8: Compute overall SMS:
total_distance_all = SUM(total_distance[i])
total_time_all = SUM(total_time[i])

SMS = total_distance_all / total_time_all

STEP 9: Convert to km/h:
SMS = SMS * 3.6

OUTPUT: SMS

Method 3: Trajectory based aggregation of Edie's Space Mean Speed

✅ Method 3: Edie's Definition (Pseudo Code + Theory)

🔷 Core Idea (Edie, 1963)

Traffic variables are defined over a **space-time region**:

- Total distance traveled:

$$\sum d_i$$

- Total time spent:

$$\sum t_i$$

🔷 Formulas

Flow (q):

$$q = \frac{\sum d_i}{L \cdot T}$$

Density (k):

$$k = \frac{\sum t_i}{L \cdot T}$$

Space Mean Speed (v):

$$v = \frac{\sum d_i}{\sum t_i}$$

Where:

- d_i = distance traveled by vehicle *i* inside region
- t_i = time spent by vehicle *i* inside region
- L = road length
- T = time interval

Method 3: Trajectory based aggregation of Edie's Space Mean Speed (contd.)

Pseudo Code:

```
INPUT:
   trajectory dataset with:
      Track_ID, Time, step_distance
   intervals = [1, 2, 5, 10, ...]
   road_length = max(cumulative_distance)
- min(cumulative_distance)

FOR each interval T in intervals:

   STEP 1: Create time bins
      For each row:
         time_bin = floor(Time / T) * T

   STEP 2: Initialize storage per time_bin
      For each time_bin:
         total_distance = 0
         total_time = 0

   STEP 3: Aggregate trajectory
contributions
      FOR each observation (row):
         total_distance += step_distance
         total_time += Δt

         NOTE:
            Δt = 1 second (since data is 1 Hz)

   STEP 4: Compute Edie variables per
time_bin

      flow (q):
         q = total_distance / (road_length *
T)

      density (k):
         k = total_time / (road_length * T)

      space_mean_speed (v):
         v = total_distance / total_time

   STEP 5: Unit conversion
      v_kmh = v * 3.6

   STEP 6: Store results per time_bin

END FOR

STEP 7: Compute summary over all time
bins

   avg_flow = mean(q)
   avg_density = mean(k)
   avg_speed = mean(v_kmh)

OUTPUT:
   - Time-bin level results
   - Interval summary results
```

Method 4: Entry-Based aggregated Count Method for Space Mean Speed Estimation (contd.)

✅ Method 4: Entry-Based Count Aggregation Space Mean Speed

🔷 Core Idea

Estimate SMS using:

- **flow from entry counts**
- **density from vehicle accumulation over space**
- then apply:

$$v_s = \frac{q}{k}$$

🧠 Mathematical Formulation

1. Flow (Entry-based)

$$q = \frac{N_{\text{entry}}}{T}$$

- N_{entry}: number of vehicles entering during interval
- T: aggregation time interval

2. Density (Space-based)

$$k = \frac{N_{\text{in segment}}}{L}$$

- $N_{\text{in segment}}$: vehicles present in road segment
- L: road length

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3. Space Mean Speed

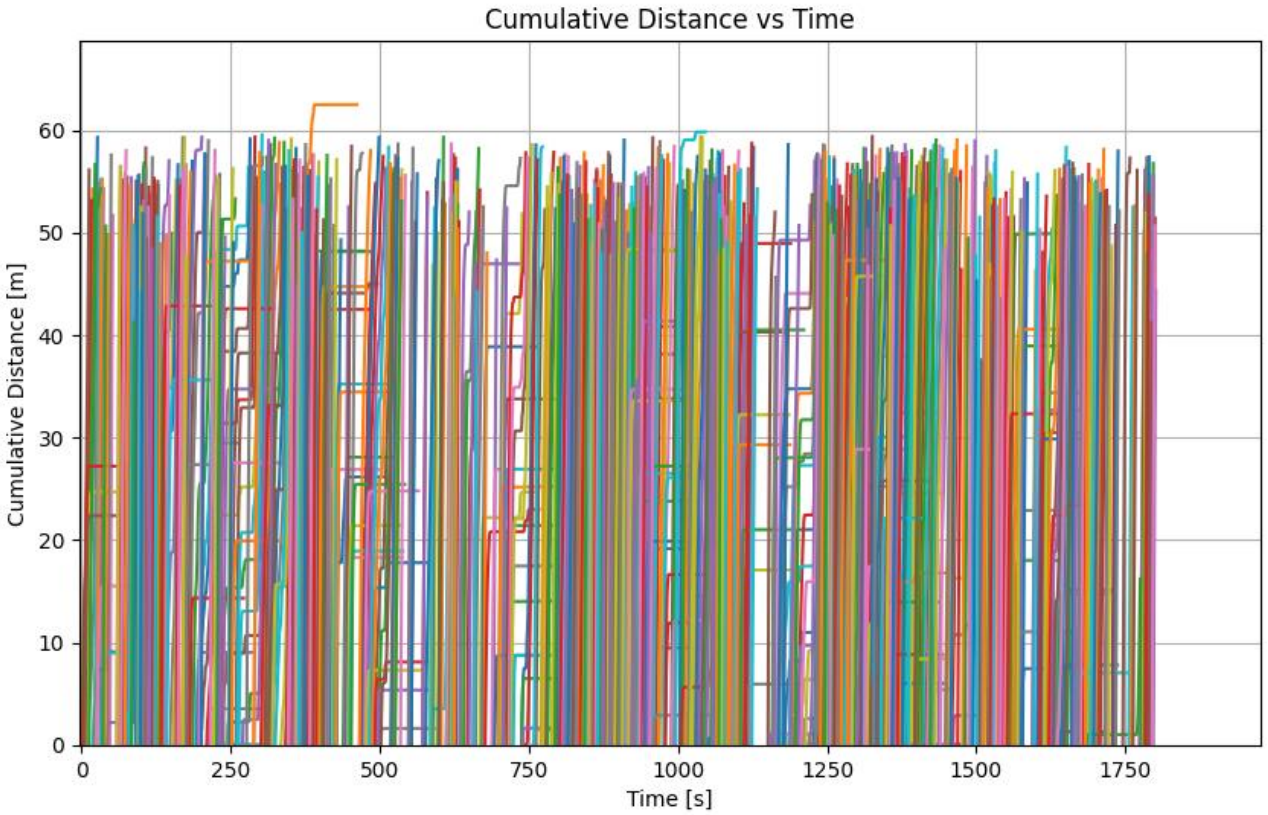
$$v_s = \frac{q}{k}$$

Method 4: Entry-Based aggregated Count Method for Space Mean Speed Estimation(contd.)

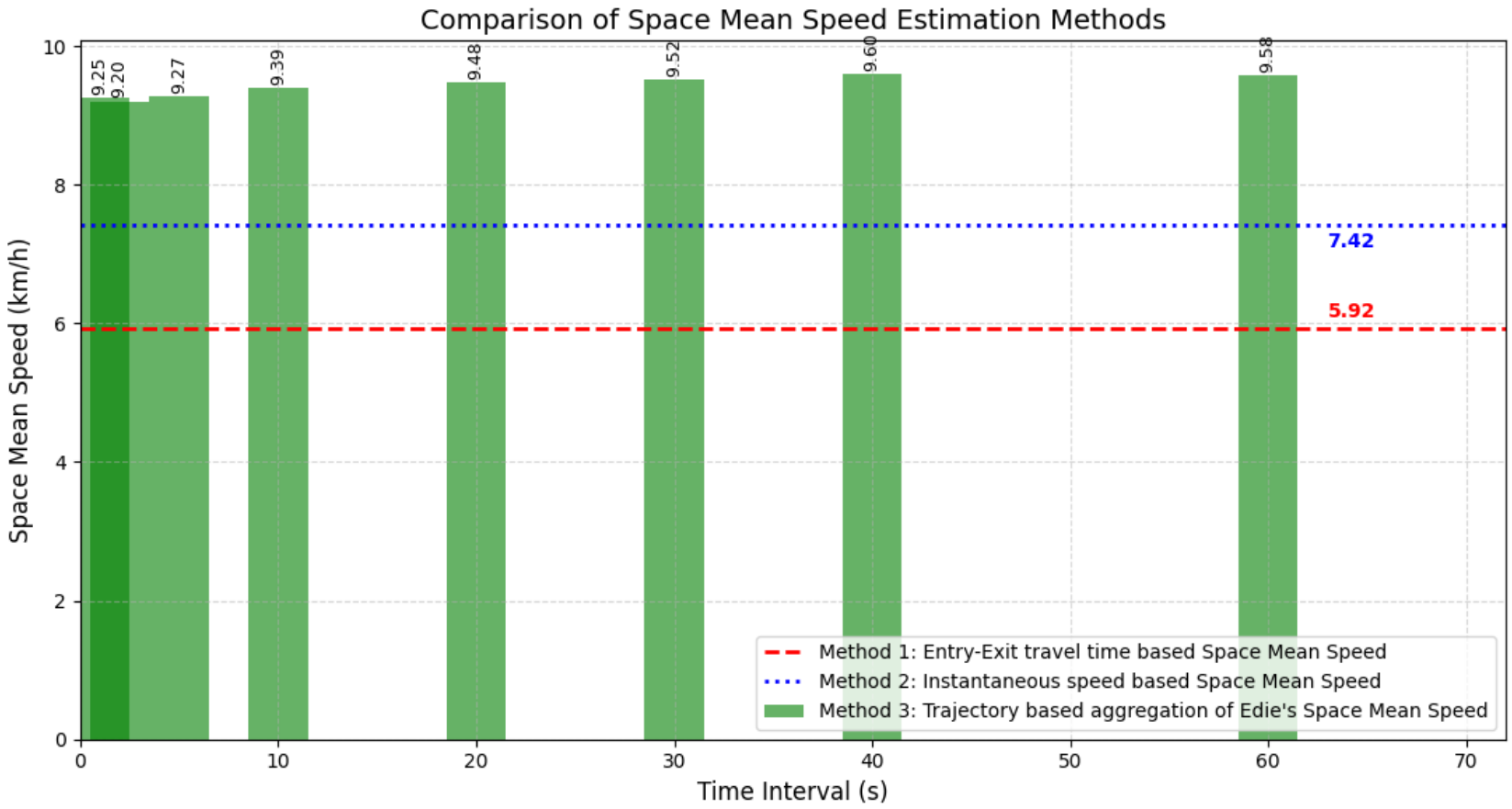
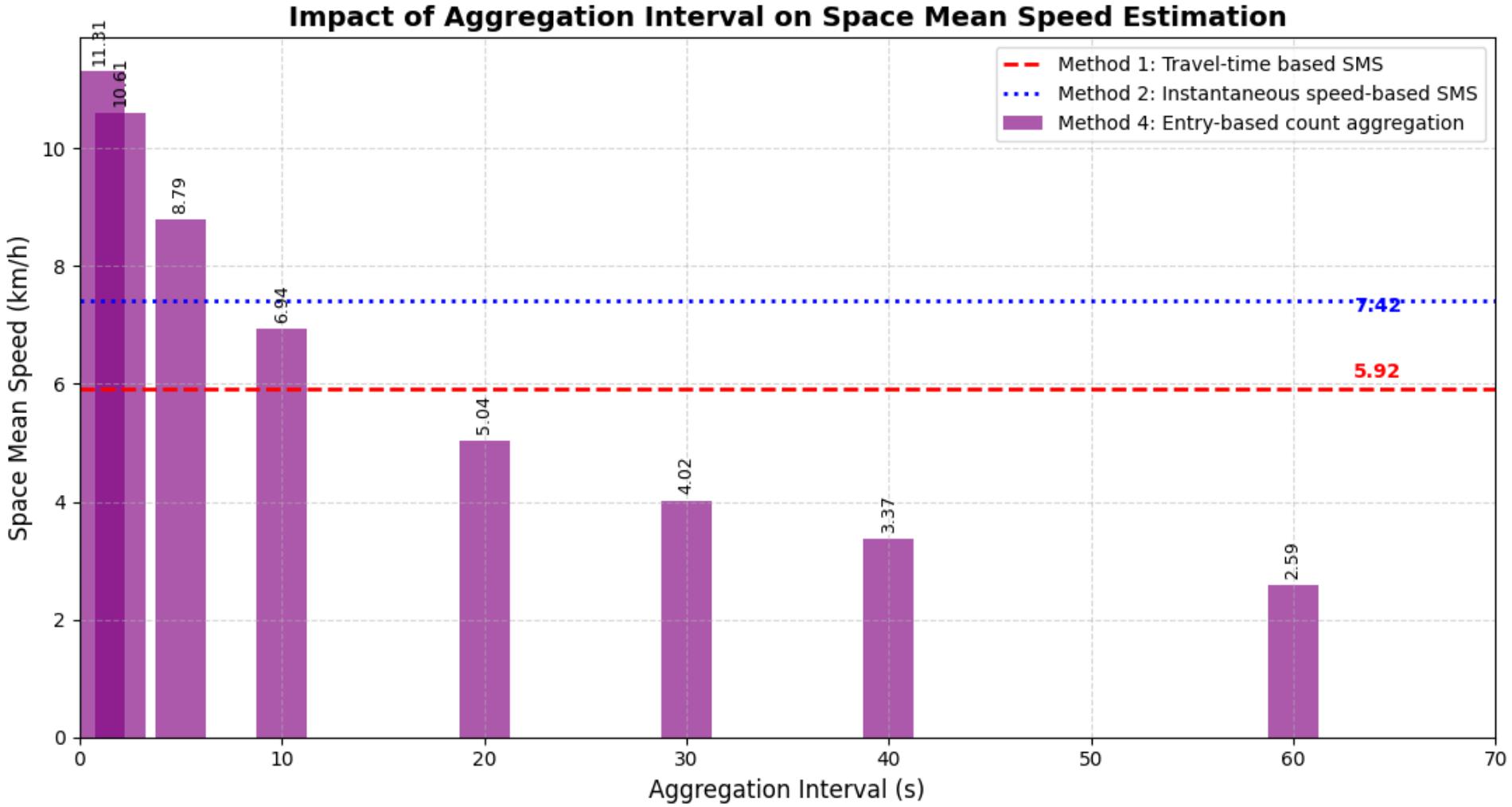
Pseudo Code:

```
INPUT:
   trajectory data (Track ID, Time, position)
   road_length L
   list of aggregation intervals (e.g., 1s, 2s, 5s,
10s)

PREPROCESS:
   Sort data by Track ID and Time
   Define study segment [x_start, x_end]

FOR each aggregation interval T:

   Divide time into windows:
      [t0, t0+T], [t0+T, t0+2T], ...

   FOR each time window W:
      ----------------------------
      STEP 1: Compute Flow (q)
      ----------------------------
      Count vehicles that ENTER segment during
W:

         N_entry = number of vehicles whose
entry time ∈ W

      Flow:
         q = N_entry / T

      ----------------------------
      STEP 2: Compute Density (k)
      ----------------------------
      For each time instant t in W:

         Count vehicles inside segment:
            N(t) = number of vehicles with position
in [x_start, x_end]

      Average number of vehicles:
         N_avg = (1 / T) * ∫ N(t) dt
            ≈ mean over sampled timestamps

      Density:
         k = N_avg / L
      ----------------------------
      STEP 3: Compute SMS
      ----------------------------
      IF k > 0:
         v_s = q / k
      ELSE:
         v_s = NaN

      ----------------------------
      STEP 4: Store Results
      ----------------------------
      Save:
         interval T
         time window W
         flow q
         density k
         space mean speed v_s

POSTPROCESS:

   For each interval T:
      Compute average SMS over all windows

OUTPUT:
   Table:
      interval | avg_flow | avg_density |
avg_space_mean_speed
```

Trajectory plots



Comparison with Edie's aggregation method



Comparison with entry-based count aggregation